

Challenge: Autonomous Line Tracking

Points: 100 • Estimated Time: 90 Minutes

Challenge Scope

The robot must autonomously navigate a 2-meter black line on a white floor without human intervention.

Instructions & Engineering Hints

- **Objective:** Design and program your autonomous robot to successfully navigate a 2-meter black line on a white floor from start to finish without human intervention.
- **Sensor Placement Tips:** Consider how the height and angle of your light or color sensors relative to the floor impact their ability to distinguish between high-contrast surfaces. Think about what happens if a sensor is mounted too high or too far off-center from the robot's turning pivot point.
- **Architectural & Structural Considerations:** Ensure your robot's chassis is balanced. Heavy overhangs can cause erratic steering or wheel slip during sharp corrections.
- **Programming Strategies:** Start by reading raw sensor values over both the white surface and the black line to establish a reliable threshold. Instead of a rigid on-off (bang-bang) control loop, consider how a proportional response might yield smoother, more continuous driving behavior along gentle curves.